

Lecture – 2

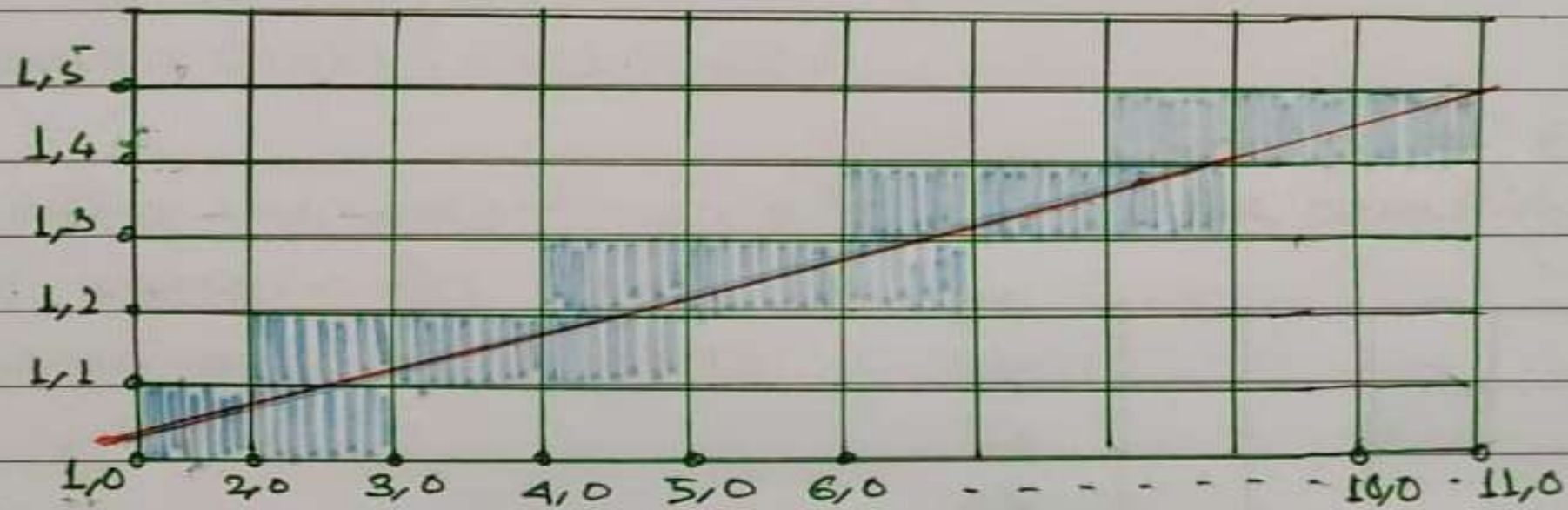
CS 372 (Computer Graphics)



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LINE DRAWING ALGORITHM

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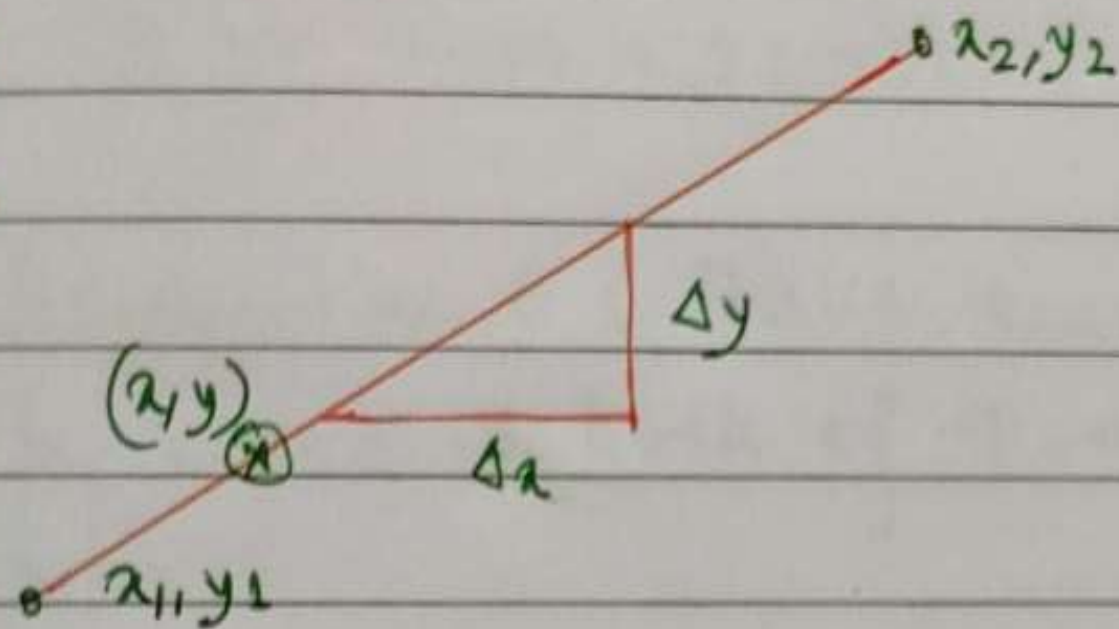
STAIRCASE EFFECT

A continuous line, will illuminate all the square pixels that falls on the line.

Line drawing algorithms help in minimizing, or choosing efficiently, the most addressable pixels.

DDA - DIGITAL DIFFERENTIAL ANALYZER

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- (*) Draw line from (x_1, y_1) to (x_2, y_2)
- (*) Compute slope $\Delta y / \Delta x$.
- (*) If (x, y) is a given pixel then next pixel $x + \epsilon \Delta x, y + \epsilon \Delta y$.

If we keep incrementing (x, y) by amounts proportional to Δx and Δy , we will get a line in the direction of (x_2, y_2) .

DDA - VALUE OF ϵ ?

Let $x = 100, y = 50$.

Case I: ϵ is very small

Let $\epsilon \Delta x = 0.001, \epsilon \Delta y = 0.002$

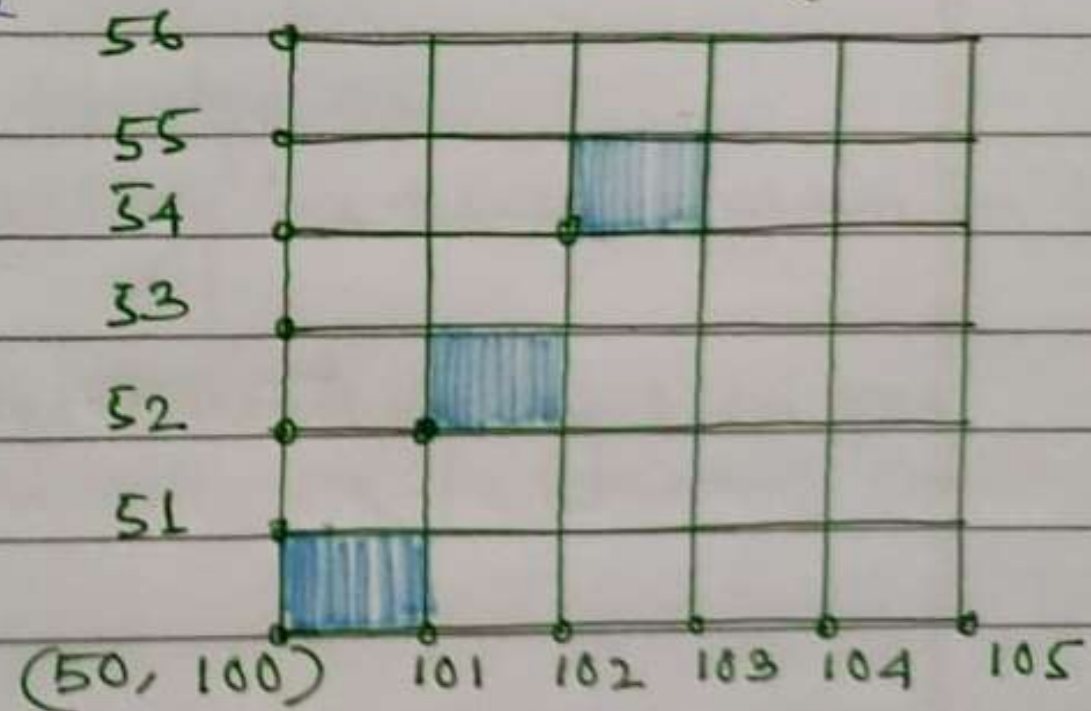
Even after incrementing, x & y will remain the same due to round-off. **Same pixel displayed.**

Case II: ϵ is very large

Let $\epsilon \Delta x = 1, \epsilon \Delta y = 2$.

Line will not be continuous.

Choice of ϵ determines the quality of DDA.



SYMMETRIC DDA

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$$\Delta x \rightarrow X_{inc} = x_2 - x_1$$

$$\Delta y \rightarrow Y_{inc} = y_2 - y_1$$

Determine ϵ ? Take X_{inc} & Y_{inc} and keep dividing by 2 until both of them are less than 1.

Example: Let $X_{inc} = 100$ and $Y_{inc} = 50$.

$$\epsilon \Delta y: \frac{50}{2} \rightarrow \frac{25}{2} \rightarrow \frac{12.5}{2} \rightarrow \frac{6.25}{2} \rightarrow \frac{3.125}{2} \rightarrow \dots \rightarrow 0.39$$

$$\epsilon \Delta x: \frac{100}{2} \rightarrow \frac{50}{2} \rightarrow \frac{25}{2} \rightarrow \frac{12.5}{2} \rightarrow \dots \rightarrow 0.78$$

Initially $x = 100, y = 50$.

$$1^{st} \text{ iter: } x = 100 + 0.78 = 100.78 \simeq 101; y = 50 + 0.39 = 50.39 \simeq 50$$

$$2^{nd} \text{ iter: } x = 100.78 + 0.78 = 101.56 \simeq \underline{102}; y = 50.39 + 0.39 = 50.78 \simeq \underline{51}$$

$$3^{rd} \text{ iter: } x = 101.56 + 0.78 = 102.34 \simeq \underline{102}; y = 50.78 + 0.39 = 51.17 \simeq \underline{51}$$

SIMPLE DDA

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Symmetric DDA : Possibility of ending up in the same pixel while incrementing in x & y direction.

Simple DDA : 1. Consider which one is greater bet. Δx & Δy .
Say $\Delta x > \Delta y$.

2. Divide Δx & Δy by Δx .

3. While incrementing we never remain in the same pixel.

We are always incrementing by 1 in one direction.

BRESENHAM'S LINE DRAWING ALGO

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DDA Disadvantage: Use of "round" function due to floating point nos. slows the algorithm.

Bresenham's algo: Given current pixel, how to choose next pixel?

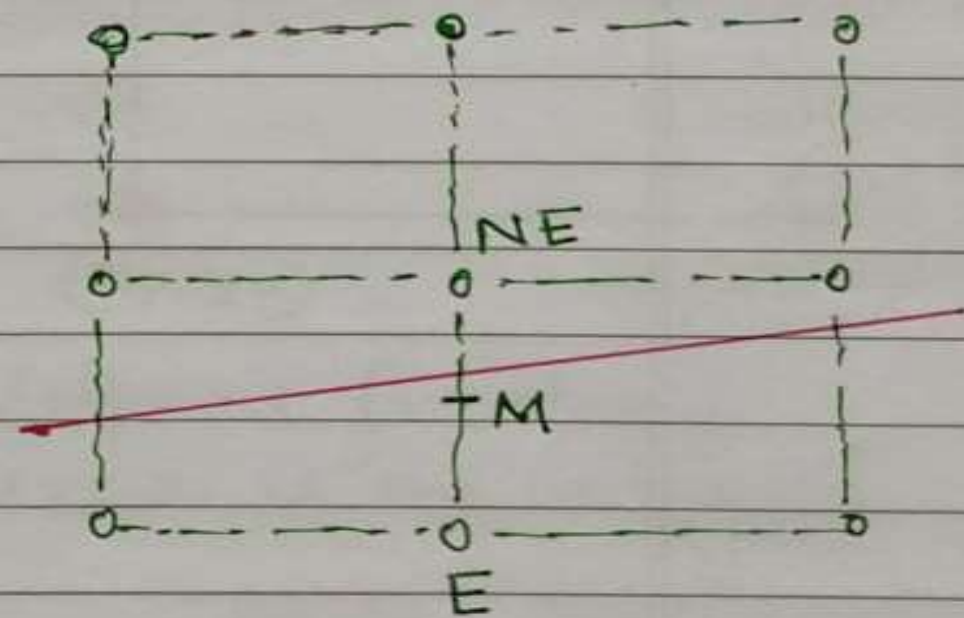
Line eqn: $y = \left(\frac{dy}{dx}\right) * x + B$

Rewrite in $F(x, y) \equiv ax + by + c = 0$

Replacing gives

$F(x, y): dy * x - dx * y + B * dx = 0$

$\Rightarrow a = dy; b = -dx; c = B * dx$



Criteria: Evaluate midpoint 'M' w.r.t the eqn of the line.

BRESENHAM CNTD..

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1. If a point lies on the line, $F(x, y) = 0$.
2. If a point is below the line, it will force $-dx * y$ to drop from the case if M would have been on the line.
→ So, the -ve quantity starts reducing, forcing F to be +ve.

o°. $F(x, y) > 0$: point M is below the line

3. $F(x, y) < 0$; M is above the line.

$$F(x, y) \equiv dy * x - dx * y + B * dx = 0$$

BRESENHAM (INTD...) 8

Here Q is above M . $F(x, y) > 0$ and we choose 'NE' as next pixel. So, which is closer to Q , 'E' or 'NE' is decided by the value of F .

Procedure:

Decision variable 'd', where $d = F(M)$.

$$= F(x_p + 1, y_p + \frac{1}{2})$$

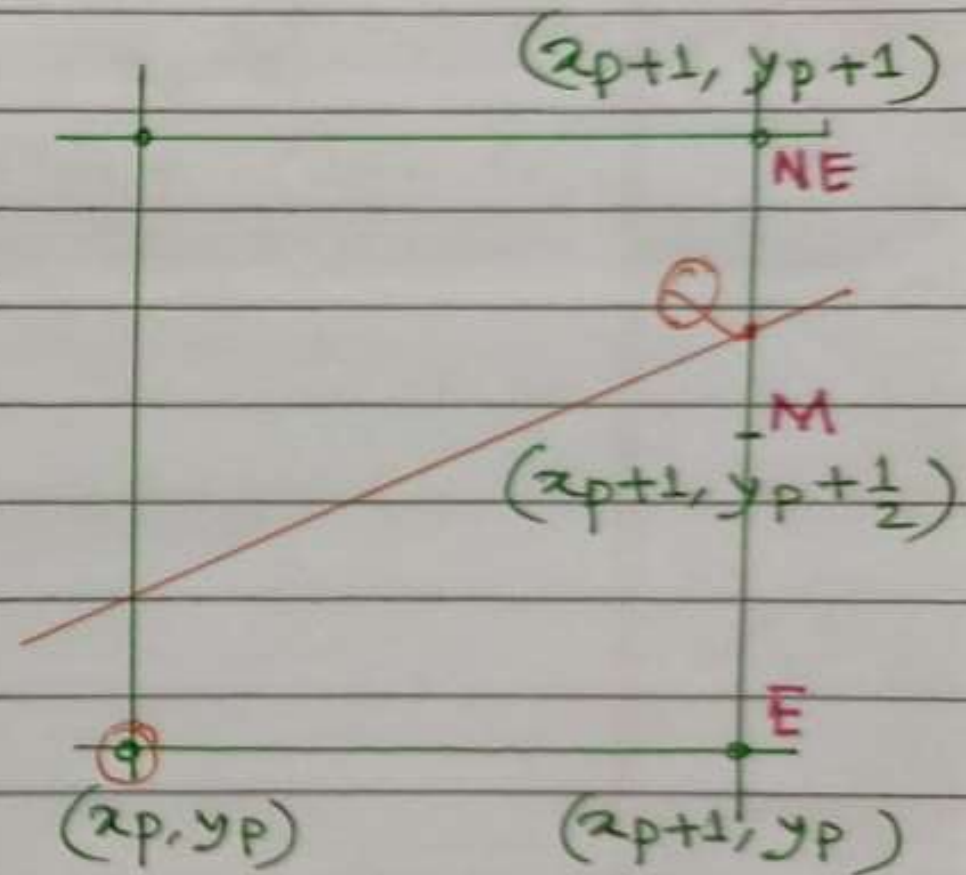
$$= a(x_p + 1) + b(y_p + \frac{1}{2}) + c.$$

Set $d_{old} = d$.

CASE 1: When 'E'

$$d_{new} = F(x_p + 2, y_p + \frac{1}{2}) \quad [\text{the next midpoint-function}]$$

$$(\Delta d)_E = d_{new} - d_{old} = a = dy \quad \{y_2 - y_1 \text{ is an integer}\}$$



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BRESENHAM (INTD...)

CASE 2: When we choose 'NE'

Next midpoint fn. will be

$$d_{\text{new}} = F(x_p + 2, y_p + \frac{3}{2}) = a(x_p + 2) + b(y_p + \frac{3}{2}) + c$$

$$(\Delta d)_{\text{NE}} = d_{\text{new}} - d_{\text{old}} = a + b = dy - dx \quad \{ \text{Again an integer} \}$$

What is d_{start} ? d @ initial position say (x_0, y_0) .

$$\begin{aligned} d_{\text{start}} &= F(x_0 + 1, y_0 + \frac{1}{2}) \quad \{ \text{midpoint of the first pixel} \} \\ &= a(x_0 + 1) + b(y_0 + \frac{1}{2}) + c \\ &= ax_0 + by_0 + c + a + \frac{b}{2} = F(x_0, y_0) + a + \frac{b}{2} \\ &= a + \frac{b}{2} = dy - \frac{dx}{2} \quad [\text{can be floating point}] \end{aligned}$$

Generalisation: Multiply the value of F by 2.

$$\therefore d_{\text{start}} = 2dy - dx, \quad (\Delta d)_E = 2dy, \quad (\Delta d)_{\text{NE}} = 2(dy - dx).$$

BRESENHAM ALGORITHM

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$$x = x_0; y = y_0; dy = y_1 - y_0; dx = x_1 - x_0$$

$$d = 2dy - dx; (\Delta d)_E = 2dy; (\Delta d)_{NE} = 2(dy - dx)$$

while ($x < x_1$)

if $d \leq 0$ /* choose E */

$$d = d + (\Delta d)_E$$

else /* choose NE */

$$d = d + (\Delta d)_{NE}$$

$$y = y + 1;$$

end if

$$x = x + 1;$$

plot - point (x, y)

end while

